

NG YI XUAN

Software Engineer

ngngvongng@gmail.com | 011-5955 8003 | [LinkedIn](#) | [GitHub](#) | Teriang, Pahang

OBJECTIVE

Final-year Computer Science student (CGPA 3.51) currently interning as Software Engineer at Godigital (Ean Label Industry). Skilled in software engineering with Laravel, React, MySQL, Docker, Redis, and AI-assisted workflows. Seeking a Software Engineer role; available full-time from September 2026. Brings 1+ year of hands-on development experience through internship and personal projects.

PROFESSIONAL EXPERIENCE

Web Developer Intern | Godigital – Ean Label Industry | Mar 2026 – Present

- Leveraged AI-assisted tools (GitHub Copilot, Gemini) to accelerate development, engineering structured prompts to guide code generation while reviewing and refining all output to ensure quality and maintainability.
- Utilized Docker and docker-compose to set up consistent local development environments, streamlining the team setup process.
- Resolved historical inventory discrepancies via complex SQL data cleaning, achieving 100% stock accuracy in the Warehouse Management System.
- Modernized legacy interfaces into minimalist dashboards using React and Nuxt.js; integrated the company ERP and warehouse systems at the database and backend level using Laravel and MySQL, keeping data synchronized across both.
- Collaborated within a 5-member team to design and build the Warehouse Management System, contributing to architecture and feature decisions.
- Proposed and implemented a Finance reporting dashboard, adopted by the team, giving the Finance department real-time visibility into stock quantity, value, movement, and aging that was previously unavailable.

PROJECTS

Web-Based Python Bootcamp for Beginners | Final Year Project | 2025 – 2026

- Built a full-stack adaptive learning platform (Laravel, React, Inertia.js, MySQL) serving lessons, coding exercises, gamification, and community features across 264 routes and 68 database migrations.
- Implemented Bayesian Knowledge Tracing to model per-concept mastery from student answers, extending it with question-type and difficulty-weighted probabilities to measure each student's learning gain against a placement baseline.
- Achieved 338 automated tests (PHPUnit, Vitest, Playwright) with CI on every push; caught and fixed a defect that silently corrupted the learning-gain baseline.
- Integrated the Gemini API for AI-assisted lesson authoring and automated concept tagging, and Judge0 for sandboxed Python execution.

Playground: Real-Time Multiplayer Game Platform | Side Project | [GitHub Repo](#) | [Live Demo](#) | 2025

- Built an extensible real-time multiplayer game platform (Draw & Guess, Werewolf) with a pluggable game-module architecture, so a new game requires only one backend module, one frontend view, and one registry line.
- Designed a server-authoritative model with per-player information hiding to prevent cheating; synced live game state across players over Socket.io, with lobby, room codes, and host controls (kick/host-transfer).
- Implemented JWT auth with guest mode and a storage layer using PostgreSQL (Supabase) in production or in-memory for local dev; built a Mahjong scoring engine supporting Hong Kong and Guobiao rules with hand decomposition and tenpai detection.

Meeting AI Platform | Side Project | [GitHub Repo](#) | 2025

- Built a full-stack AI meeting platform using Laravel that transcribes meetings with OpenAI Whisper and generates summaries, key topics, and action items using Gemini AI.
- Replaced polling with real-time updates via Laravel Reverb (WebSocket); used Redis for caching and as the broadcast/queue driver to improve responsiveness under load.
- Containerized the full stack with Docker and docker-compose (Laravel, Redis, and database) for consistent, reproducible environments; implemented chunked file upload (5 MB slices) for large audio files.
- Set up a CI pipeline with GitHub Actions running 62 automated feature tests on every push, ensuring code quality and preventing regressions.

EDUCATION

BSc Computer Science (Hons) – Network Engineering | Universiti Malaysia Sabah | Expected Nov 2026

CGPA: 3.51 | Internship: Mar – Aug 2026 | Available full-time from Sep 2026

Relevant Coursework: Network Programming, Cybersecurity, Web Technology, Cloud Computing, Big Data Analytics, Object-Oriented Programming, System Analysis & Design

TECHNICAL SKILLS

Frontend: HTML5, CSS3, JavaScript, TypeScript, React, Nuxt.js, Inertia.js

Backend: PHP, Laravel, Java, C++, REST API

Database: MySQL, SQL, Redis

Concepts: REST API, WebSocket (Laravel Reverb), Automated Testing

AI Tools: GitHub Copilot, Gemini AI, OpenAI Whisper, Prompt Engineering

Tools: Docker, Git, Figma

Languages: Malay (Native), English (Fluent), Mandarin (Fluent)

CERTIFICATIONS

Huawei HCIA – Datacom & Cloud Computing | CCNA: Enterprise Networking, Security & Automation

CCNA: Switching, Routing & Wireless Essentials | CCNA: Introduction to Networks